

ESP32 Motion-Activated Audio Player Wiring Layout

Components

- ESP32 Dev Board
- SD Card Module (SPI)
- IR Motion Sensor (digital output)
- Push Button
- Audio Amplifier
- Speaker

YELUFT Micro SD TF Adapter Card Reader → ESP32

SD Module Pin	ESP32 Pin
CS	GPIO 5
SCK	GPIO 18
MOSI	GPIO 23
MISO	GPIO 19
VCC	5V
GND	GND

IR Motion Sensor → ESP32

IR Sensor Pin	ESP32 Pin
OUT (signal)	GPIO 15
VCC	3.3V
GND	GND

Button → ESP32

Button Pin	ESP32 Pin
One leg	GPIO 26
Other leg	GND

ESP32 → PAM8302A Amplifier

ESP32	Amplifier
GPIO 25	Audio IN+
GND	Audio IN-
VDC	3.3V
GND	GND
Shutdown	3.3V

Amplifier → Speaker

Amplifier Pin	Connects To
Speaker OUT (L/R)	Speaker terminal

This is for converting audio to WAV

<https://audio.online-convert.com/convert-to-wav>

Change bit depth 8-bit

Change audio sample rate 16000Hz

Change audio channels Mono

Normalize audio none

Change PCM format no change